

PROJECT TITLE:

Digital Passwird System with 3 wrong attempts

SUBJECT: Digiatl Logic Design

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Digital Lock System with 3 wrong attempts Counter

Abstract

This project demonstrates a 4-bit digital lock system designed by using

blub,wires and basic logic gates which is the

- X-OR Gate
- AND Gate
- NOT Gate

The system verifies a **4-bit code** entered by the user against a pre-defined code stored in the back-end. If the user entered code match with the stored code, the system activates the **first load (e.g., Bulb D1)** for the first stage. A secondary condition can activate another **load (e.g., Bulb D2)** for advanced control. This lock is an example of how combinations of logic gates can be applied for secure access control without micro controllers.

Introduction

A digital lock system is an access control mechanism used to permit entry only when a correct code is entered.

In this project, logic gates are used to compare the entered 4-bit binary code with a pre-set code.

First of all, if all the inputs match with the pre-set code exactly, then system activates an output device as a result then lock is deactivate. Another condition is, if the users enter code does not match with preset code then the second output device is automatic activate which is the act like a security alarm. The second output device enhance the security of digital lock system.

This design uses AND, X-OR, and NOT gates to implement the logic comparison and control. The lock system works purely on combinational logic and does not require memory or software, making it suitable for low-cost, hardware-based security systems.

Objective

To design a 4-bit digital lock system using logic gates. Compare two sets of binary inputs and display the result using two different LEDs which is represent two conditions as like lock is open, lock is activate.

If all inputs are equal to pre-set code, then green LED will turn ON; if any input is different frm pre-set code, the red LED will turn ON.

To demonstrate a two-stage unlocking system (D1 and D2) for enhanced security.

Components Used

In this digital lock system the following components are used

1. Logic Gates:

- **XOR Gates:**
  XOR Gates (U1, U2, U3 and U4) is used for bit wise comparison between two input lines.
- **NOT Gates:**
  NOT Gates (U5, U6, U7, U8 and U14) is used to invert the XOR output or control signals.
- **AND Gates:**
  AND Gates (U9, U10, U11, U12 and U13) is used to combine signals for final LED control.

2. **Inputs:**

- **Binary Input Switches:**

Represented by red and blue switches. Red switch represents (logic 1) while the blue switches represents (logic 0) switches.

There are 8 switches in total to provide input combinations for the digital lock system.

3. **Outputs:**

There are only two outputs for the digital lock system which is behave different in different conditions.

- **Condition 1:**

  A certain logic condition is met GREEN-LED (D1) Lights up.
- **Condition 2:**

  When the opposite condition is met RED-LED (D2) Lights up

4. Ground Connection:

Ground (Earth) connection is used as the common reference point for the circuit's voltage.

5. Wires/Connections:

Green lines (Wires) represent the electrical connections between gates and components.

Working Principle

1. The user enters a 4-bit code using the switches.
2. Each input is fed into the logic gate network where:
   - X-OR gates first process the inputs according to the required code pattern.
   - NOT gates invert signals where the stored code requires a logic '0'.
   - AND gates combine and verify that all bits match the pre-defined code.
3. If the entered code matches exactly:

Then there are two stages

Stage 1: D1 Green bulb turns ON, indicating correct code.

Stage 2: A secondary AND and NOT combination checks for another condition before turning ON **D2 Red** bulb.

Wrong Attempts Counter (Security Extension)

Objective

To further enhance the security of the digital lock, a "Wrong Attempt Counter" has been integrated. This circuit tracks the number of incorrect password entries. If the user enters the wrong code three times, the system provides visual feedback via a 7-segment display and triggers a security alert (Red LED).

Components Added

- **IC 7476 (Dual JK Flip-Flop):** Used to construct a 2-bit asynchronous counter to store the number of wrong attempts.
- **IC 7447 (BCD to 7-Segment Decoder):** Converts the binary count from the flip-flops into a format readable on the display.
- **7-Segment Display:** Visually shows the number of incorrect attempts (0, 1, 2, 3).
- **Logic Gates:** Used to detect the "wrong code" condition and generate the clock pulse for the counter.

Working Principle

1. Detection of Wrong Code:

   Whenever the user enters a code that does not match the preset password, the main comparator circuit generates a specific logic signal (Logic '1'). This signal acts as a Clock Pulse for the counter circuit.

2. Counting Mechanism (U13:A & U13:B):

   The circuit uses two JK Flip-Flops (U13:A and U13:B) configured as a 2-bit Ripple Up-Counter.

   - The **J and K inputs** are set to Logic High (1), putting the flip-flops in "toggle mode."
   - On every incorrect attempt, the "Wrong Code" signal triggers the clock input of the first flip-flop.
   - The output of the first flip-flop feeds into the second, allowing the circuit to count in binary: 00 (0), 01 (1), 10 (2), and 11 (3).

3. Displaying the Count (U14):

   The binary outputs from the flip-flops are connected to the A and B inputs of the 7447 Decoder (U14). The 7447 IC processes this binary data and illuminates the corresponding segments on the 7-segment display. This allows the user to see exactly how many failed attempts have occurred.

4. Security Trigger (Lockout):

   When the counter reaches the 3rd wrong attempt (Binary 11), the Red LED (D1) is activated. This serves as a warning or a system lockout, preventing further attempts until the system is manually reset.

Circuit Connections Description

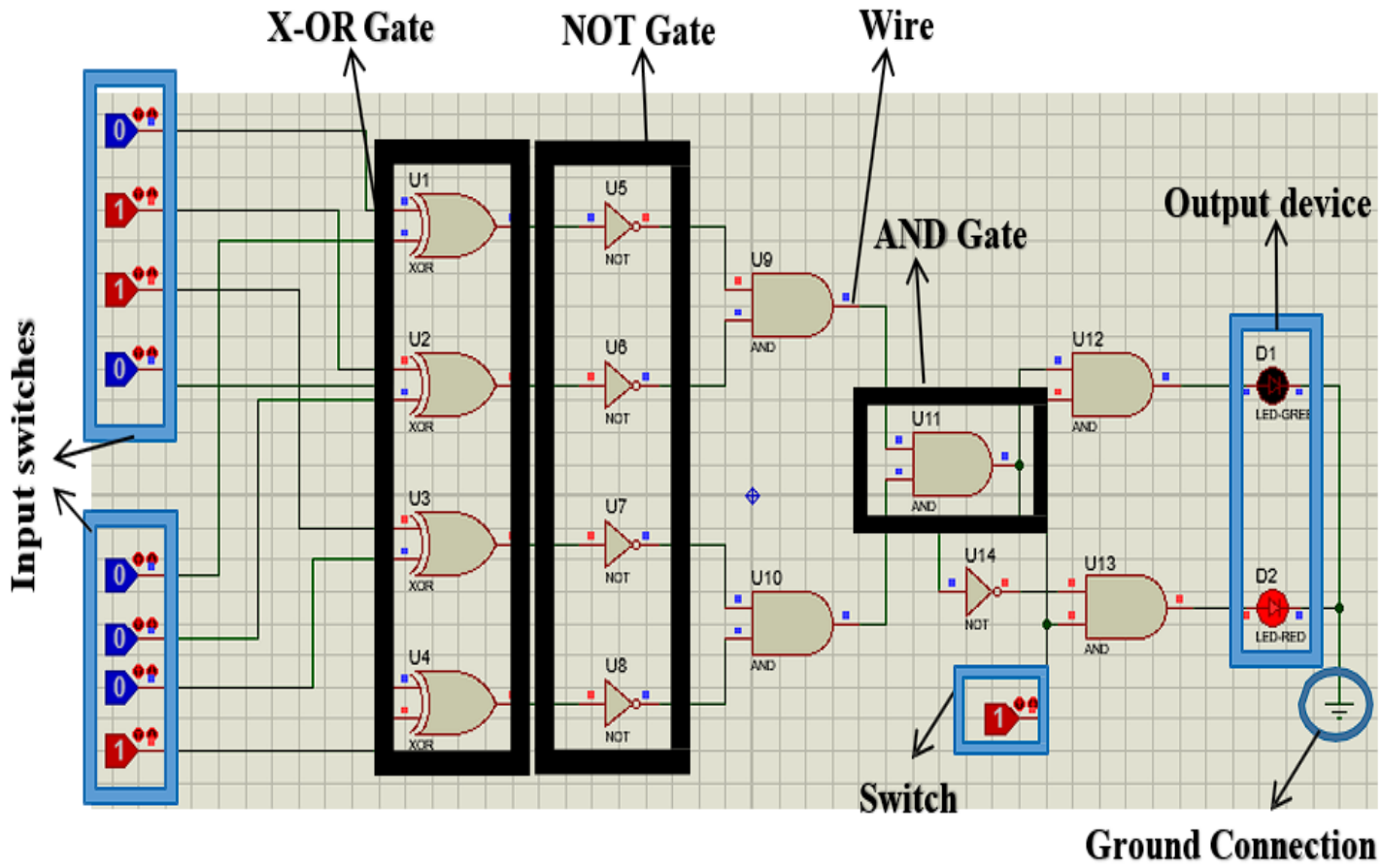
- The **Clock Input** of the first JK Flip-Flop (U13:A) is connected to the output of the digital lock's "Mismatch" logic (where the entered code $\neq$ stored code).
- The **Q Output** of U13:A is connected to the Clock of U13:B to form the counter chain.
- The **Q Outputs** of both flip-flops connect to the inputs of the U14 (7447) decoder to drive the display.

Truth Table for counter:

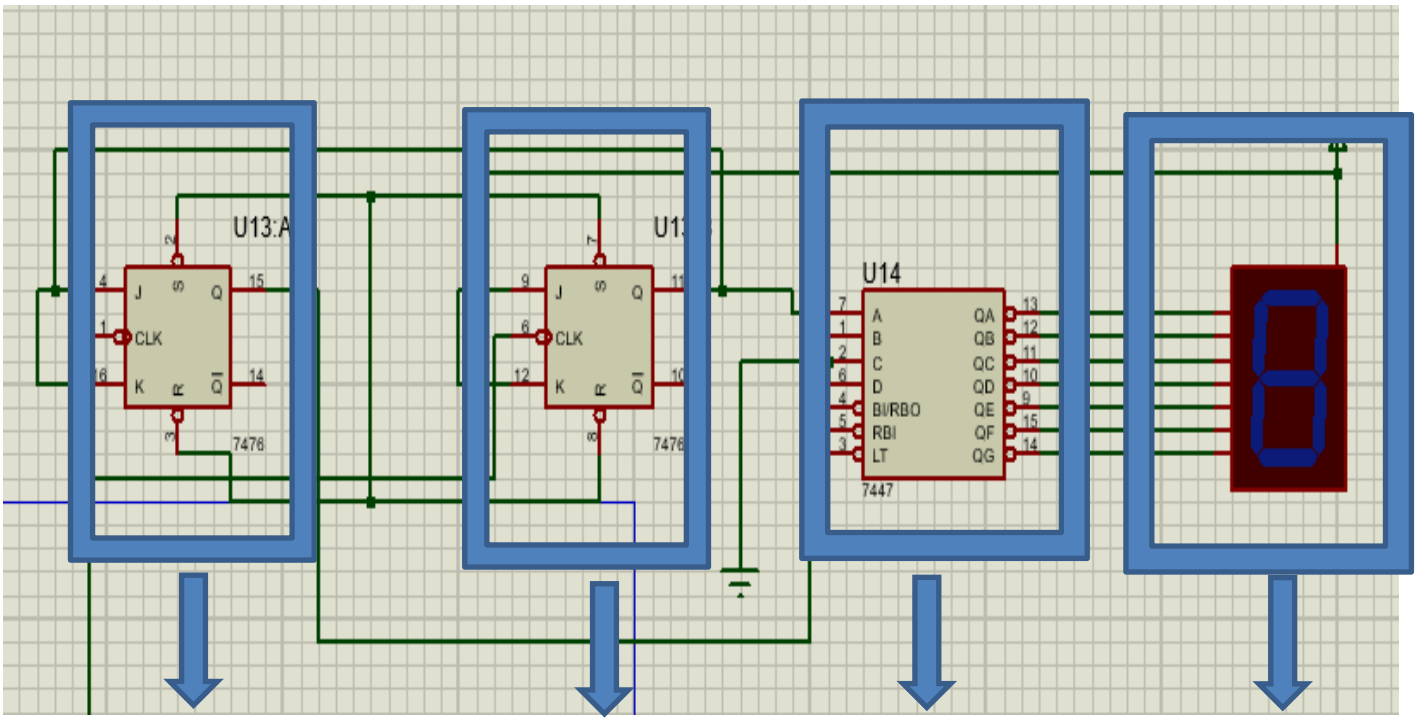
Truth Table: 3 Wrong Attempts Counter

Attempt Status	Clock Pulse (Wrong Input)	Flip-Flop Output (Q_1)	Flip-Flop Output (Q_0)	Decimal Count	7-Segment Display	System Status
Initial State	0	0	0	0	**0**	Secure / Normal Operation
1st Wrong Attempt	1 (Pulse)	0	1	1	**1**	Warning - 1st Strike
2nd Wrong Attempt	1 (Pulse)	1	0	2	**2**	Warning - 2nd Strike
3rd Wrong Attempt	1 (Pulse)	1	1	3	**3**	**ALARM TRIGGERED (Red Light ON)**

Circuit Description:



Counter Circuit for showing wrong attempts



First JK FLIP FLOP	Second JK FLIP FLOP	7447 BCD IC	7Segment Display

- Inputs (A, B, C, D) represent the four bits entered by the user.
- The stored code is embedded in the way inputs are connected through NOT gates.
- X-OR gates (U1 to U4) are used to create match conditions for each bit.
- NOT gates (U5 to U8) invert the output of XOR gates as per requirement.
- AND gates (U9 to U11) check for simultaneous correctness of all bits.
- **D1 Green** bulb glows when the first condition is met.
- Additional gates (U13) verify secondary unlocking before D2 is turned ON.

Applications

- ✓ Home and Office Digital Locks:
  For securing doors.
- ✓ Electronic Safes :
  To protect jewelry or valuables.
- ✓ Restricted Area Entry Control:
  For laboratories, server rooms, or secure zones.
- ✓ School/University Locker Systems :
  To safeguard students' belongings.
- ✓ Small Hardware Security Systems:
  Where micro controllers or software are not required.

Truth table with off password switches:

Main password				User enter password				Pass word switch	Output devices	
A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**X**	**GREEN**	**RED**
0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	0	0	0
0	0	1	0	0	0	1	0	0	0	0
0	0	1	1	0	0	1	1	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	0	1	0	1	0	1	0	0	0
0	1	1	0	0	1	1	0	0	0	0
0	1	1	1	0	1	1	1	0	0	0
1	0	0	0	1	0	0	0	0	0	0
1	0	0	1	1	0	0	1	0	0	0
1	0	1	0	1	0	1	0	0	0	0
1	0	1	1	1	0	1	1	0	0	0
1	1	0	0	1	1	0	0	0	0	0
1	1	0	1	1	1	0	1	0	0	0
1	1	1	0	1	1	1	0	0	0	0
1	1	1	1	1	1	1	1	0	0	0

In the given table the password switch is off, so the out-device does not perform any function. When the switch is off, User code is match with the preset code or not the output devices will not perform their functions.

Truth table with ON password switches:

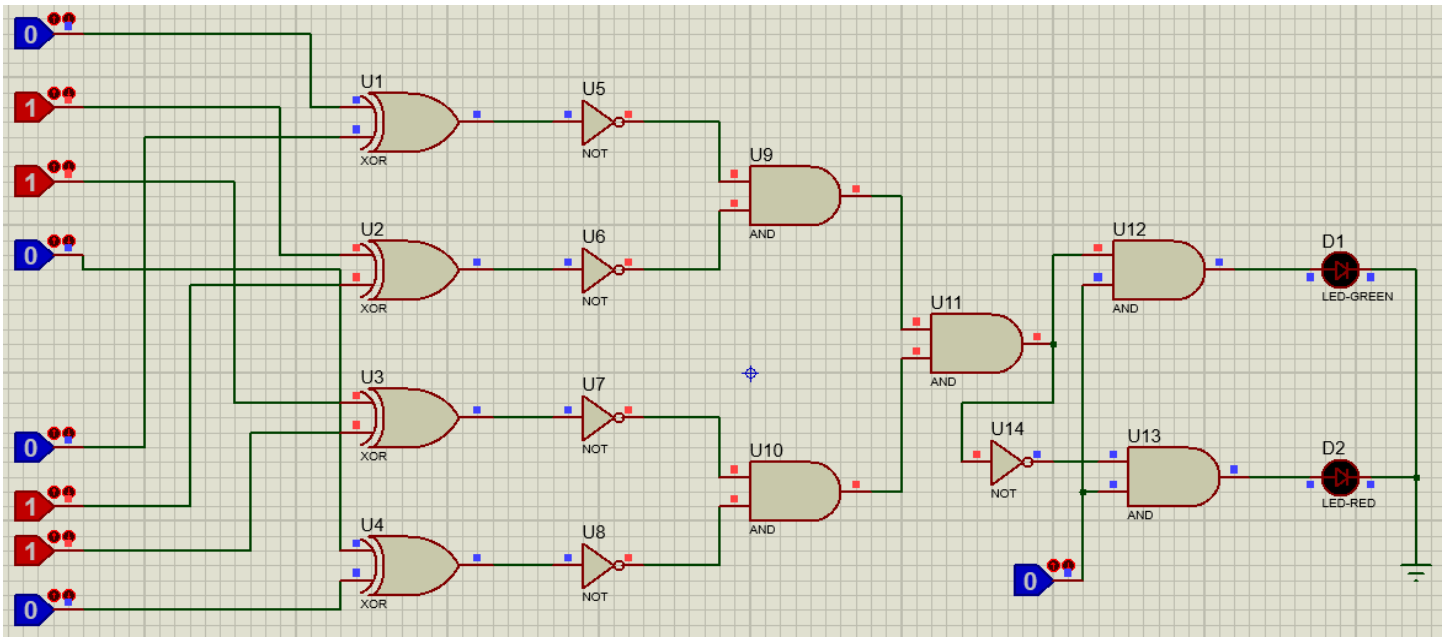
Main password				User enter password				Pass word switch	Out put devices	
A	B	C	D	E	F	G	H	X	GREEN	RED
0	0	0	0	0	1	0	0	1	0	1
0	0	0	1	0	0	0	1	1	1	0
0	0	1	0	0	0	1	0	1	1	0
0	0	1	1	0	0	1	1	1	1	0
0	1	0	0	0	1	0	0	1	1	0
0	1	0	1	0	1	0	0	1	0	1
0	1	1	0	0	1	1	0	1	1	0
0	1	1	1	0	1	1	1	1	1	0
1	0	0	0	1	0	0	0	1	1	0
1	0	0	1	1	0	1	1	1	0	1
1	0	1	0	1	0	1	0	1	1	0
1	0	1	1	1	0	1	1	1	1	0
1	1	0	0	1	1	1	0	1	0	1
1	1	0	1	1	1	0	1	1	1	0
1	1	1	0	1	1	1	0	1	1	0
1	1	1	1	1	1	0	1	1	0	1

In the given table the password switch is ON, so the out-device preform their function.

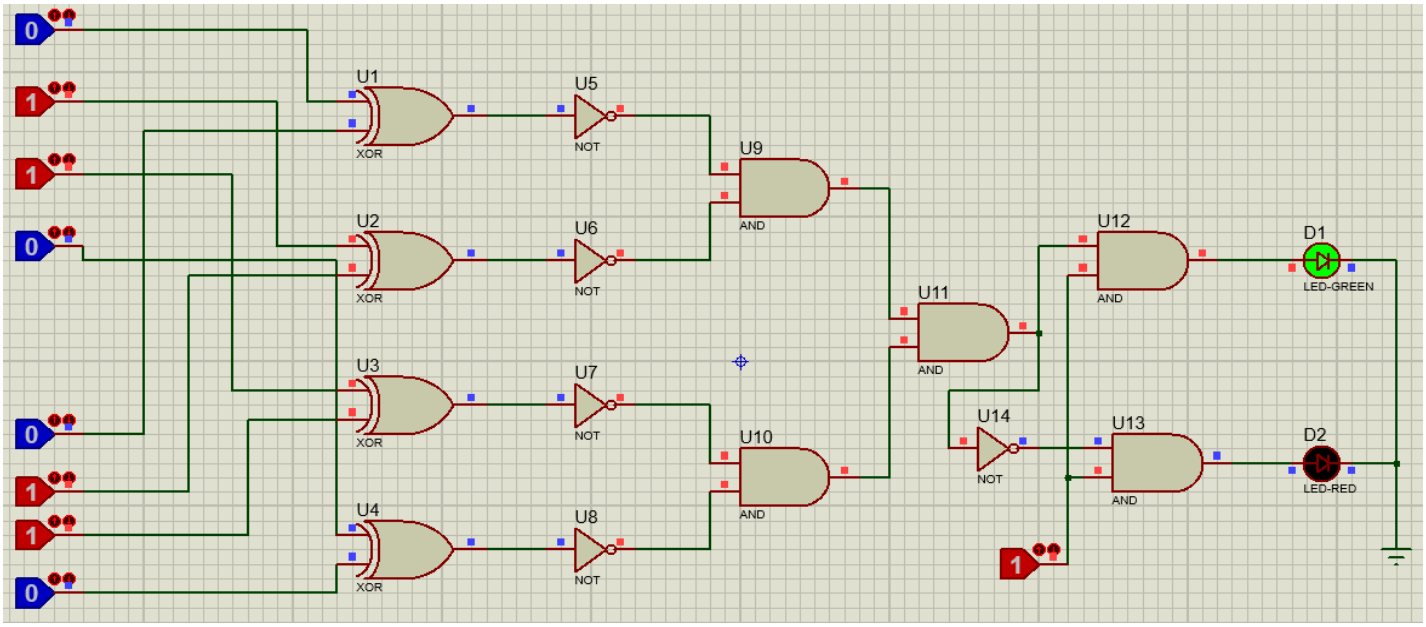
When the switch is ON and user code is match with the preset code only when devices will perform their functions.

Simulation Details

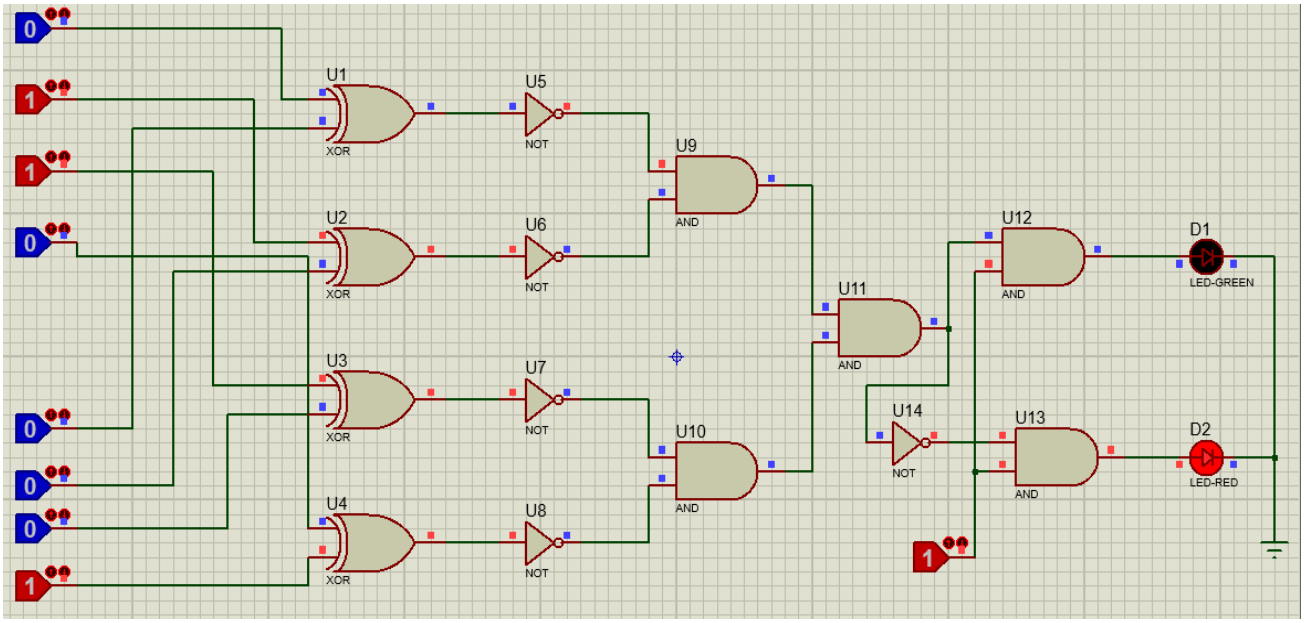
This digital lock system is design by using Proteus software.



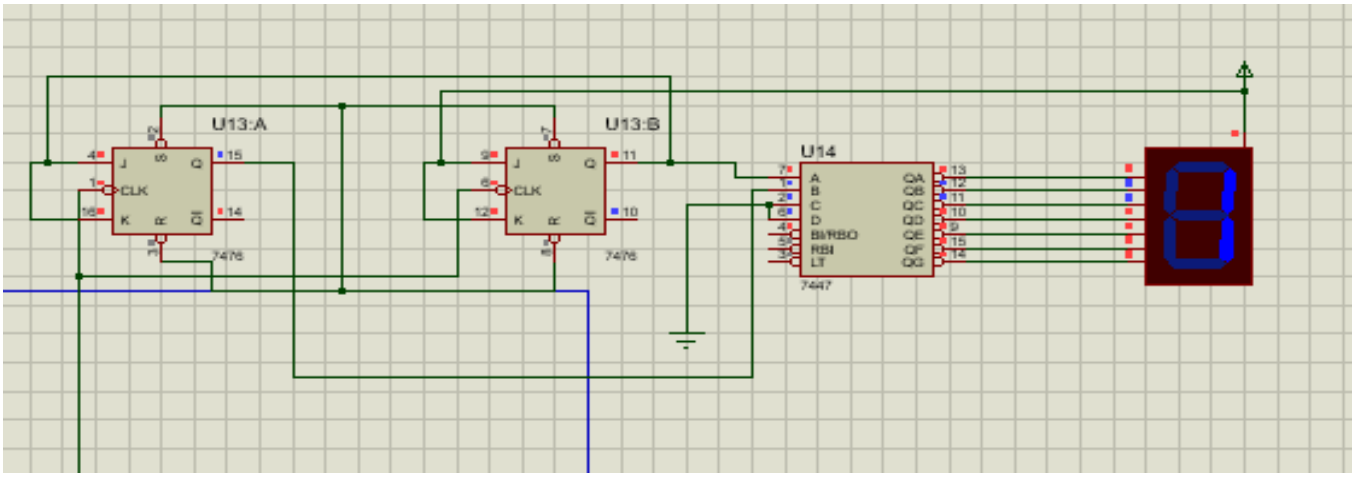
Circuit with on switch and green light



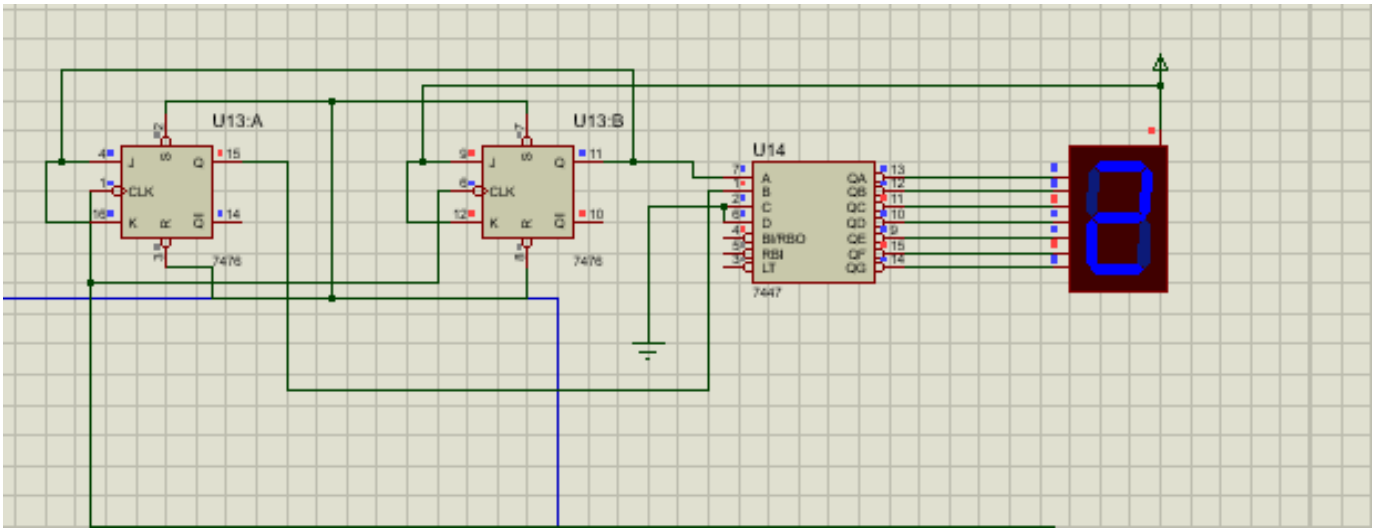
Circuit with on switch and red light



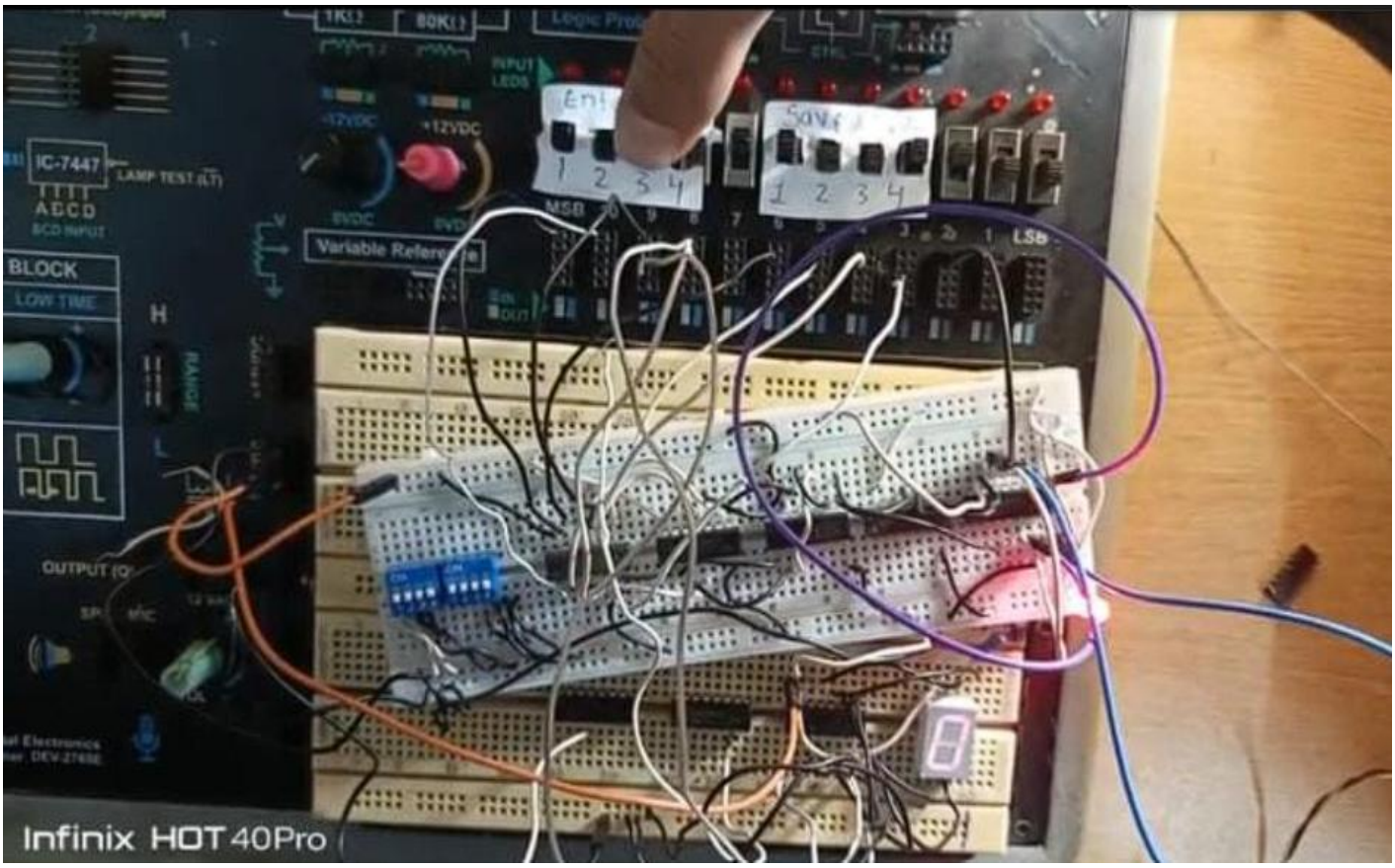
Circuit with first wrong attempt

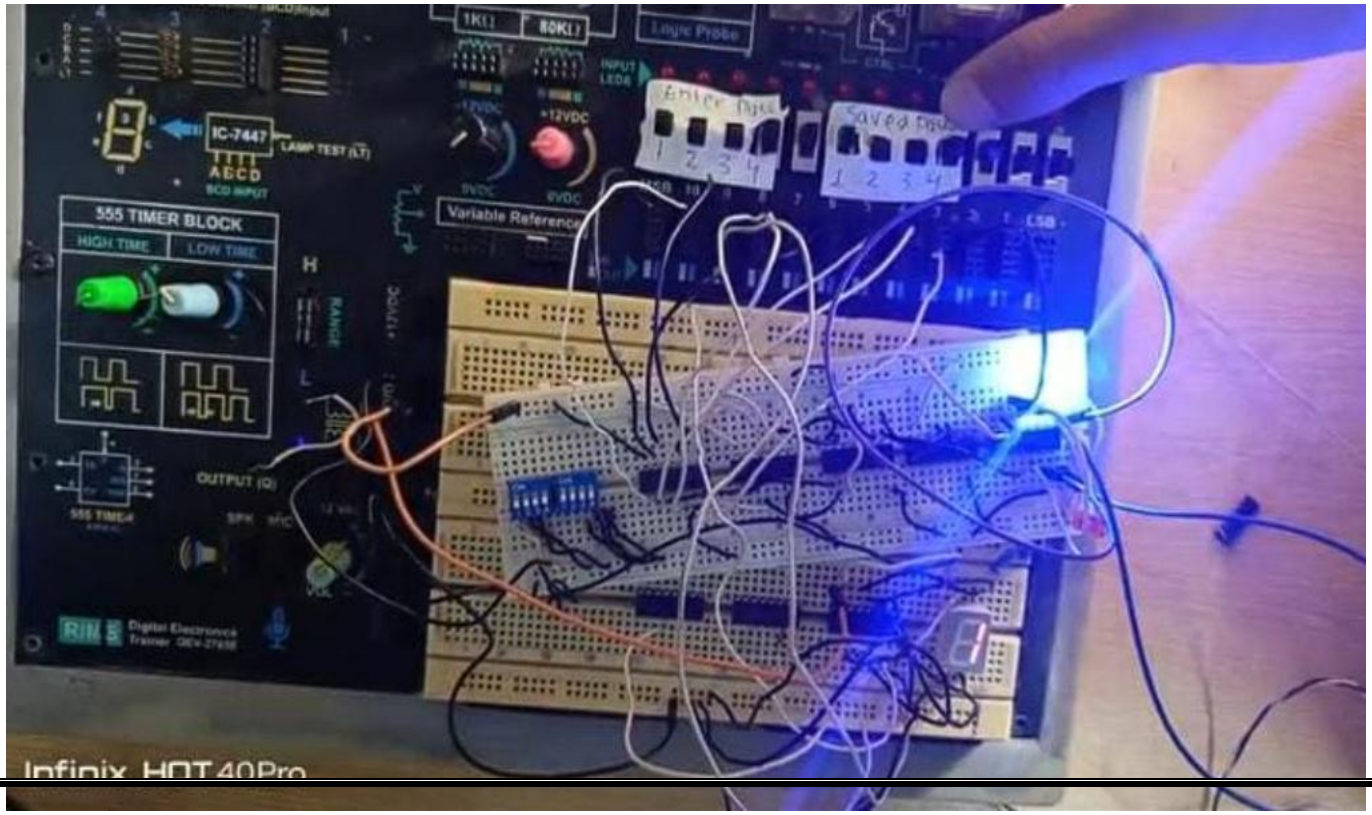


With 2nd wrong attempt



Practical Simulation





Limitations

- ✓ Used only fixed code instead of programmable code.
- ✓ Limited to 4-bit inputs (16 combinations).
- ✓ Use ground connection to prevent short circuit.

Possible Improvements:

- ✓ Using 6-bit or 8-bit codes for more security.
- ✓ Use an alarm which activate on wrong code.
- ✓ Adding a buzzer for incorrect attempts.
- ✓ Using flip-flops for code sequence entry.
- ✓ Adding a time delay to prevent brute-force attempts.

Safety Precautions

- ✓ Avoid short-circuits when wiring.
- ✓ Ensure correct polarity for LEDs.

Advantages

- ➢ Low Cost:
  No expensive micro controller as result its cost is very low.
- ➢ No Programming Needed :
  Fully hardware-based system there is no need of any program to activate lock.
- ➢ Simple Design and Assembly:
  Uses basic components and wiring for the digital code system.
- ➢ Easy Modification :
  The preset code can be changed by rewiring the input switches.
- ➢ Fast Response:
  Being combinations of logic-based, it works without processing delay, as result our response is very fast.

Conclusion:

This project successfully demonstrated the design and simulation of a 4-bit digital lock system using basic logic gates instead of complex microcontrollers. The circuit effectively compares a user-entered binary code against a preset password stored in the hardware. It reliably provides immediate visual feedback, turning on a green LED for a successful entry and a red LED for an incorrect one. This part of the project established a functional foundation for hardware-based security logic.

To significantly enhance security, a critical feature was added: a "wrong attempt counter" that tracks failures from 0 to 3. Using JK flip-flops and a 7-segment display, the system now visually counts every incorrect password entry. Upon reaching the third consecutive wrong attempt, the system triggers a persistent alarm state. This addition is vital because it stops users from endlessly guessing the password, making the digital lock far more secure and practical for real-world applications.